

FairCDR: Transferring Fairness and User Preferences for Cross-Domain Recommendation

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Abstract

Cross-domain recommendation (CDR) has gained significant attention for its ability to address data sparsity issue. However, most existing CDR methods focus primarily on improving recommendation accuracy while largely overlooking fairness considerations, which can lead to biased outcomes and unfair treatment of different user groups. To solve this critical problem, we investigate whether fairness can be transferred from the source domain to the target domain. Our analysis suggests that fairness can be effectively transferred if the fairness of the source domain is ensured and the distributions of the source and target domains are well aligned. Based on this, we propose the **FairCDR**, a novel framework that can achieve the knowledge transfer of fairness and user preferences simultaneously. FairCDR owns two phases: single-domain fairness guarantee and inter-domain distribution alignment. In the first phase, we employ an adversarial learning-based recommender (**ALR**) to disentangle user preferences from sensitive attributes in the source domain. In the second phase, we introduce a new mutual learning-based diffusion model (**MLDiff**), which engages in mutual learning with ALR to progressively align the distributions of the source and target domains. This improves ALR's adaptability

to distribution shifts, ultimately ensuring fairness and recommendation performance in the target domain. Extensive experiments on multiple real-world cross-domain datasets demonstrate that FairCDR surpasses existing strong baselines in both fairness and recommendation quality.

CCS Concepts

• **Information systems** → **Information retrieval**.

Keywords

Cross-domain recommendation, Fairness transferability, Diffusion models

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1 Introduction

In recent years, recommender systems (RS) have been successfully applied in many modern digital platforms, significantly enhancing user experience by providing personalized services [11, 23–25]. A long-standing challenge in RS is how to handle data sparsity issue where the user-item interactions are very sparse. Cross-domain recommendation (CDR) provides a principled way to alleviate this data sparsity issue and has gained widespread attention [2, 39]. The core idea of CDR is transferring the shared user preferences

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learned from the source domain to the target domain for improving recommendation performance.

An increasing number of CDR approaches have been introduced lately, and they have demonstrated promising results [2, 13]. For example, CoNet [13] employs a cross-connection neural network to transfer user preferences between two domains, thereby improving recommendation results for the target domain. Although expressive, these approaches focus solely on enhancing recommendation accuracy, neglecting the issue of *fairness* in the target domain. In fact, fairness is an especially critical issue, since unfair recommendation outcomes are not only unethical but also detrimental to the long-term interests of RS itself [4, 41]. However, most existing recommenders have only focused on ensuring fairness within a single domain, guaranteeing fairness in CDR is an essential yet largely under-explored topic.

An ideal CDR model should not only achieve good recommendation performance but also ensure fairness. This is highly challenging because the user-item interactions in the target domain are often limited. Directly imposing existing fairness approaches [20, 38] on user representations may further weaken the valuable user information in the target domain, making it difficult to accurately capture user preferences, which can ultimately lead to a significant decline in recommendation quality. Similar to the transfer of user preferences in CDR, a natural question arises: *Can we leverage the abundant data resources in the source domain to effectively transfer fairness to the target domain?*

In this paper, we systematically investigate this important question. We first analyze the upper bound of the fairness distance in the target domain by quantifying the gap between the source and target domains and identifying its influencing factors. Fairness distance [33] measures the bias in a model’s fairness, which increases when predictions vary significantly due to sensitive features. Our analysis reveals that this upper bound is closely related to the extent of distribution alignment between domains: the more thoroughly the distributions are aligned, the smaller the fairness distance gap becomes. This suggests that fairness can be effectively transferred if the fairness of the source domain is ensured and the distributions of the source and target domains are well aligned. Based on this analysis, we then propose the **FairCDR**, a novel framework that simultaneously transfers fairness and user preferences.

Specifically, FairCDR consists of two phases: **Single-domain Fairness Guarantee** and **Inter-domain Distribution Alignment**. In the first phase, we train a fairness-aware single-domain recommender in a manner of adversarial learning, i.e., **ALR**, which can remove the influence of sensitive attributes from user representations in the source domain. In the second phase, we introduce a new mutual learning-based diffusion model, i.e., **MLDiff**, to progressively align the distributions of the source and target domains, facilitating more effective knowledge transfer of fairness and user preferences. MLDiff first learns the distributional prior of the target domain. Unlike common diffusion models [12, 36], MLDiff utilizes this prior during the reverse process, enabling it to generate transitional distributions along the direction of the target domain distribution, to align the source domain with the target domain in K diffusion processes. MLDiff and ALR mutually learn during training. At each diffusion process, MLDiff first produces

the transitional distribution guided by the target domain distribution; ALR is then trained by both the transitional distribution and the source domain distribution, allowing it to effectively capture fairness and user preferences in the target domain. In general, the contributions of this paper are summarized as follows,

- We theoretically study whether and how fairness can be transferred in CDR, with the goal of building an ideal CDR model that can ensure fairness in the target domain while maintaining strong recommendation performance.
- We propose the FairCDR, a novel framework which utilizes MLDiff to progressively align the source and target domain distributions during the diffusion process. This enables ALR to continuously perceive distributional changes, thereby enhancing its capability of guaranteeing fairness and capturing user preferences in the target domain.
- We conduct extensive experiments on multiple real-world cross-domain datasets. Empirical results and model analyses show that FairCDR achieves the superior performance in both fairness and recommendation quality¹.

2 Related Works

Fairness in RS. Existing fairness solutions in RS are broadly classified into three categories: pre-processing, in-processing, and post-processing methods. Pre-processing methods, e.g., fair sampling and data balancing [7, 17] aim to mitigate bias in the data prior. In-processing methods address bias during the model training, which can be categorized into two types: fairness-constrained and *adversarial learning*-based approaches. The former imposes fairness as an additional criterion, applying hard constraints to ensure that the solutions fall within a fair region [22]. The latter uses the generator to generate representations and the discriminator to predict sensitive attributes [38, 44]. When the sensitive attributes can no longer be effectively predicted, fairness is considered to have been achieved. Adversarial learning-based approaches are widely used in fairness algorithms in RS, where the adversarial training between the generator and discriminator can reduce the influence of sensitive attributes effectively. Post-processing methods focus on modifying the generated recommendation outcomes to improve fairness [29]. However, most current methods focus on fairness in single-domain RS, while our work focuses on fairness in CDR.

Cross-Domain Recommendation. CDR aims to enhance recommendation quality in the target domain by utilizing user-item interaction data gathered from other related source domains. Existing CDR approaches are broadly classified into two categories: *embedding-and-mapping* and *joint training* methods. Embedding-and-mapping methods try to map users or items from both the source and target domains into a shared latent space, or transferring representations from the source domain to the target domain through a mapping function [46]. Joint training-based methods aim to simultaneously learn cross-domain representations, allowing the model to better capture shared information between domains while minimizing the influence of domain-specific details [3, 43]. But current CDR methods only emphasize improving model performance,

¹The source code is available at <https://github.com/aabbbcd/FairCDR>.

often overlooking user fairness in the target domain, which is the critical issue this paper aims to address.

Diffusion Models. Diffusion models are a class of latent variable generative model based on probabilistic diffusion process [12, 32]. Diffusion models simulate a forward diffusion process during training by incrementally adding noise to the data, and then reverse this process during generation by denoising the data, allowing for the generation of high-quality samples [12]. In recent years, diffusion models have achieved significant success in various domains, such as computer vision [5, 32] and natural language processing [34]. Recent study has begun to explore the potentials of diffusion models in RS. For example, Wang et al. [36] use diffusion models to simulate and learn the generation process of user interactions, thereby predicting future interaction probabilities between users and items. Different from above methods, our work develop a new diffusion model to progressively align the distributions of the source and target domains, enabling the fairness-aware recommendation model trained on the source domain to effectively adapt to the target domain.

3 Problem Setup

CDR Task. We first provide a description of CDR. CDR usually includes two domains: the source domain $\mathcal{S}$ and the target domain $\mathcal{T}$. $\mathcal{D}^{\mathcal{S}} = (\mathcal{U}^{\mathcal{S}}, \mathcal{V}^{\mathcal{S}}, \mathcal{E}^{\mathcal{S}}, \mathcal{X})$ and $\mathcal{D}^{\mathcal{T}} = (\mathcal{U}^{\mathcal{T}}, \mathcal{V}^{\mathcal{T}}, \mathcal{E}^{\mathcal{T}}, \mathcal{X})$ denote the data of $\mathcal{S}$ and $\mathcal{T}$, respectively. Take $\mathcal{D}^{\mathcal{S}}$ as an example, $\mathcal{U}^{\mathcal{S}}$ and $\mathcal{V}^{\mathcal{S}}$ are user and item sets in $\mathcal{S}$, $\mathcal{E}^{\mathcal{S}}$ denotes the corresponding interactions between users and items. The same description also applies to $\mathcal{D}^{\mathcal{T}}$. $\mathcal{X}$ represents the user attributes shared by $\mathcal{S}$ and $\mathcal{T}$, which can be divided into the sensitive attributes $\tilde{\mathcal{A}}$ and the insensitive attributes $\mathcal{A}$. Additionally, there are a small portion of overlapping users, i.e., $\mathcal{U}^{\mathcal{O}} = \mathcal{U}^{\mathcal{S}} \cap \mathcal{U}^{\mathcal{T}}$. Typically, $\mathcal{E}^{\mathcal{T}}$ is quite scarce, so the goal of CDR is to transfer user preferences learned from $\mathcal{S}$ to $\mathcal{T}$, using $\mathcal{U}^{\mathcal{O}}$ as a shareable bridge for enhancing recommendation performance on $\mathcal{T}$. The used notations are summarized in Appendix A.

Counterfactual Fairness. It aims to ensure decision-making models treat individuals fairly by considering hypothetical scenarios [16]. A model satisfies counterfactual fairness if, for any individual, the model's decision outcome remains the same when a sensitive attribute, such as race or gender is hypothetically changed, while keeping all other relevant variables unchanged. This approach evaluates fairness by comparing the actual outcome with the potential outcome in a hypothetical scenario where the sensitive attribute is altered, thereby preventing these attributes from having an undue influence on the decision-making process. Here we provide a formal definition of counterfactual fairness in RS.

Definition 1 (Counterfactual fairness in RS [20]). A recommendation model is considered to satisfy counterfactual fairness if, for any user u with the insensitive attribute $\mathcal{A} = \mathbf{a}$ and the sensitive attribute $\tilde{\mathcal{A}} = \tilde{\mathbf{a}}$, the Top- N recommendation list $\ell(u)$ remains unchanged when the sensitive attribute $\tilde{\mathcal{A}}$ is hypothetically replaced by any alternative value $\tilde{\mathbf{a}}'$:

$$P\left(\ell_{\tilde{\mathcal{A}} \leftarrow \tilde{\mathbf{a}}}(u) \mid \mathcal{A} = \mathbf{a}, \tilde{\mathcal{A}} = \tilde{\mathbf{a}}\right) = P\left(\ell_{\tilde{\mathcal{A}} \leftarrow \tilde{\mathbf{a}}'}(u) \mid \mathcal{A} = \mathbf{a}, \tilde{\mathcal{A}} = \tilde{\mathbf{a}}\right),$$

where $\tilde{\mathbf{a}}$ and $\tilde{\mathbf{a}}'$ represent the concrete values of $\tilde{\mathcal{A}}$ in the actual and counterfactual scenarios, respectively.

Highlights. Our work concentrates on counterfactual fairness for users because it is a prominent individual-level fairness objective [16]. In the context of RS, many studies have explored achieving better counterfactual fairness to ensure individual-level fairness while maintaining personalized recommendation performance in a single domain [14, 20, 45], while we study whether and how counterfactual fairness can be transferred in CDR.

4 Methodology

To investigate the transferability of fairness in CDR, we first conduct a theoretical analysis. We then propose the FairCDR to transfer fairness while maintaining recommendation quality.

4.1 Theoretical Analysis

In this section we provide a theoretical analysis of the transferability of counterfactual fairness in CDR. Specifically, drawing on the concept of fairness distance from the equality of opportunity framework [10, 33], we propose a novel notion termed as counterfactual fairness distance.

Definition 2 (Counterfactual fairness distance). Let G be a prediction (recommendation) model and $G(u)$ is the Top- N recommendation list for the user u , counterfactual fairness distance of G can be described as

$$d(G) \triangleq \sum_{u \in \mathcal{U}} |\mathbb{E}[G_{\tilde{\mathcal{A}} \leftarrow \tilde{\mathbf{a}}}(u \mid \mathcal{A} = \mathbf{a}, \tilde{\mathcal{A}} = \tilde{\mathbf{a}})] - \mathbb{E}[G_{\tilde{\mathcal{A}} \leftarrow \tilde{\mathbf{a}}'}(u \mid \mathcal{A} = \mathbf{a}, \tilde{\mathcal{A}} = \tilde{\mathbf{a}})]|.$$

It reflects the difference in outcomes generated by G for each user u in the actual scenario ($\tilde{\mathcal{A}} \leftarrow \tilde{\mathbf{a}}$) and the counterfactual scenario ($\tilde{\mathcal{A}} \leftarrow \tilde{\mathbf{a}}'$), thereby measuring how far G is from achieving counterfactual fairness perfectly.

The gap between the source and the target domains in terms of counterfactual fairness distance is defined as

$$\epsilon = d^{\mathcal{T}}(G) - d^{\mathcal{S}}(G), \quad (1)$$

where $d^{\mathcal{S}}(G)$ and $d^{\mathcal{T}}(G)$ represent the counterfactual fairness distances for the source and target domains, respectively. Subsequently, our objective is to establish an upper bound of ϵ . Specifically, we derive the following theorem:

Theorem 4.1. Let $\mathcal{H}$ be a hypothesis space of VC dimension d . If $\mathcal{U}^{\mathcal{S}\tilde{\mathcal{A}}}$ and $\mathcal{U}^{\mathcal{T}\tilde{\mathcal{A}}}$ are sample sets of size m' , each drawn respectively from the distributions $\mathcal{D}^{\mathcal{S}}$ and $\mathcal{D}^{\mathcal{T}}$ in the source and target domains, then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$ (over the choice of samples), for all $G \in \mathcal{H}$ (where $\mathcal{H}$ is a symmetric hypothesis space), the counterfactual fairness distance ϵ (see Eq.(1)) in the target domain is bounded above by

$$\epsilon \leq \frac{1}{2} d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{U}^{\mathcal{S}\tilde{\mathcal{A}}}, \mathcal{U}^{\mathcal{T}\tilde{\mathcal{A}}}) + \pi_H(\mathcal{U}^{\mathcal{S}\tilde{\mathcal{A}}}, \mathcal{U}^{\mathcal{T}\tilde{\mathcal{A}}}) + \lambda^*,$$

where $\lambda^* = \lambda(d, m', \delta) + \lambda_{\mathcal{A}}(G^*, f)$, $\lambda(d, m', \delta) = 8\sqrt{\frac{2d \log(2m') + \log(\frac{2}{\delta})}{m'}}$, π_H denote the Hausdorff distance and $\lambda_{\mathcal{A}}(G^*, f) = \epsilon_{\mathcal{A}}^{\mathcal{T}}(G^*, f) + \epsilon_{\mathcal{A}}^{\mathcal{S}}(G^*, f)$. The first two terms on the RHS of the inequality are

both related to the inter-domain distribution distance. Different from existing works [33], Theorem 4.1 provides an upper bound of the fairness distance in terms of counterfactual fairness, and the concrete proof can be found in Appendix B.

Through the above analysis, we found that the upper bound of ϵ depends on the extent of distribution alignment between $\mathcal{S}$ and $\mathcal{T}$. A higher degree of alignment corresponds to a smaller discrepancy in the counterfactual fairness distance between $\mathcal{S}$ and $\mathcal{T}$. Therefore, to guarantee fairness in $\mathcal{T}$, it is necessary to ensure counterfactual fairness in $\mathcal{S}$ and the alignment between domain distributions.

4.2 FairCDR

We introduce the FairCDR which achieves the knowledge transfer of fairness and user preferences to guarantee fairness and improve recommendation performance on the target domain simultaneously. Based on the above analysis in Section 4.1, FairCDR includes two phases: single-domain fairness guarantee and inter-domain distribution alignment. The whole training framework of FairCDR is given in Figure 1.

4.2.1 Single-domain Fairness Guarantee. To guarantee fairness on the source domain, we introduce the adversarial learning-based recommender (ALR) which employs adversarial learning (AL) techniques [38, 44] to mitigate biases related to sensitive attributes in user representations. By minimizing the influence of sensitive attributes, ALR ensures that recommendation outcomes remain consistent when sensitive attributes are altered, thereby achieving counterfactual fairness. Specifically, ALR first constructs an item-item collaboration graph $\mathcal{G}^S$ in the source domain $\mathcal{S}$, where each node represents an individual item. If two items are interacted with by the same user, an edge is created between these two items, and the edge weight is calculated based on the number of common interacting users. The adjacency matrix of $\mathcal{G}^S$ is given as $\mathbf{A}^S \in \mathbb{R}^{|\mathcal{V}^S| \times |\mathcal{V}^S|}$, where each element can be defined as

$$\mathbf{A}^S(v_i, v_j) = \begin{cases} \frac{|\mathcal{U}_i^S \cap \mathcal{U}_j^S|}{\sqrt{|\mathcal{U}_i^S| \cdot |\mathcal{U}_j^S|}} & \text{if } |\mathcal{U}_i^S \cap \mathcal{U}_j^S| > 0 \\ 0 & \text{otherwise,} \end{cases} \quad (2)$$

where $\mathcal{U}_i^S$ and $\mathcal{U}_j^S$ are the sets of users who have interacted with items v_i and v_j , respectively.

ALR then utilizes the GCN [15] to encode $\mathcal{G}^S$ for generating item representations in the $(l+1)$ -th layer:

$$\mathbf{Z}^{S,(l+1)} = \sigma \left((\mathbf{D}^S)^{-\frac{1}{2}} \mathbf{A}^S (\mathbf{D}^S)^{-\frac{1}{2}} \mathbf{Z}^{S,(l)} \mathbf{W}^{(l)} \right), \quad (3)$$

where σ denotes the activation function, i.e., $\text{ReLU}(\cdot)$, $\mathbf{D}^S$ is the degree matrix of $\mathbf{A}^S$, $\mathbf{Z}^{S,(l)}$ represents the item representations at the l -th layer and $\mathbf{W}^{(l)}$ denotes the model parameters. We denote the L -th layer representation $\mathbf{Z}^{S,(L)}$ as the final item representations $\mathbf{Z}_{\mathcal{V}}^S$. Building upon that, ALR generates the corresponding user representation via aggregating the interacted item representations:

$$\mathbf{Z}_u^S = \text{AGGREGATE} \left(\left\{ \mathbf{Z}_v^S \mid v \in \mathcal{I}^S(u) \right\} \right), \quad (4)$$

where $\mathcal{I}^S(u)$ represents the set of items interacted with the user u .

To achieve the fairness-aware recommendation on $\mathcal{S}$, ALR includes a generator G and a discriminator Γ . Based on $\mathbf{Z}_{\mathcal{U}}^S$, G is

used to produce unbiased user representations, ensuring that the recommendation results are not influenced by sensitive attributes. In contrast, Γ aims to infer the user's sensitive attributes based on G 's outputs. The adversarial learning between G and Γ can be formulated as

$$\mathcal{L}_{\text{adv}} = \frac{1}{|\mathcal{U}^S|} \sum_{u \in \mathcal{U}^S} [\tilde{a} \log \Gamma(G(\mathbf{Z}_u^S)) + (1 - \tilde{a}) \log(1 - \Gamma(G(\mathbf{Z}_u^S)))], \quad (5)$$

where $\tilde{a}$ denotes the binary value of the sensitive attributes, $\mathbf{Z}_u^S$ and $G(\mathbf{Z}_u^S)$ respectively represent the user representation and the unbiased user representation for u , and $\Gamma(G(\mathbf{Z}_u^S))$ is the predicted probability of u 's sensitive attribute. Γ aims to maximize Eq.(5) to improve its detection ability, while G minimizes Eq.(5) to reduce the influence of sensitive attributes in $\mathbf{Z}_u^S$. The total loss function for training ALR is given as

$$\mathcal{L}_{\text{ALR}} = \mathcal{L}_{\text{rec}} + \gamma \mathcal{L}_{\text{adv}}. \quad (6)$$

Here $\mathcal{L}_{\text{rec}}$ represents a recommendation loss produced by an arbitrary base recommendation model working on $G(\mathbf{Z}_u^S)$ and $\mathbf{Z}_{\mathcal{V}}^S$ (In Section 5.1.4, we illustrate the concrete implementation of the base recommendation model.). γ is a harmonic factor between $\mathcal{L}_{\text{rec}}$ and $\mathcal{L}_{\text{adv}}$. By optimizing Eq.(6), ALR can simultaneously satisfy fairness guarantee and personalized recommendation on $\mathcal{S}$.

4.2.2 Inter-domain Distribution Alignment. When fairness on $\mathcal{S}$ is ensured, inter-domain distribution alignment is required to transfer the learned fairness and user preferences from $\mathcal{S}$ to $\mathcal{T}$. In our work, we introduce a new diffusion model termed as mutual learning-based diffusion (MLDiff) to achieve this goal. MLDiff follows a progressive manner for aligning the distribution of the source domain P^S with the distribution of the target domain P^T . Compared with previous alignment methods in RS [6, 26], MLDiff can avoid the loss of semantic information caused by direct alignment, facilitating a better knowledge transfer of fairness and user preferences.

Due to the data sparsity of the target domain, accurately characterizing P^T is inaccessible. Hence, MLDiff introduces P_θ to approximate P^T , and the learned P_{θ^*} can serve as a distribution prior to better guide the alignment of P^S with P^T . To be specific, MLDiff first constructs an item-item collaboration graph $\mathcal{G}^T$ to learn user representations $\mathbf{Z}_{\mathcal{U}}^T$ and item representations $\mathbf{Z}_{\mathcal{V}}^T$ in the same way as Eq.(2-4). Then, the learning process of P_θ can be given here:

① *Forward Process.* The initial state is denoted as $\mathbf{X}_0 \sim P^T(\mathbf{X}_0)$ and $\mathbf{X}_0 = \mathbf{Z}^T = [\mathbf{Z}_{\mathcal{U}}^T, \mathbf{Z}_{\mathcal{V}}^T]$. The forward process gradually degrades $\mathbf{X}_0$ by adding Gaussian noise at each step from 0 to T , which can be formulated as

$$Q(\mathbf{X}_t | \mathbf{X}_{t-1}) = \mathcal{N}(\mathbf{X}_t; \sqrt{1 - \beta_t} \mathbf{X}_{t-1}, \beta_t \mathbf{I}), \quad (7)$$

where $\beta_t \in (0, 1)$ controls the noise scale at step t and $\mathbf{I}$ is an identity matrix. By applying the reparameterization trick [12] and the additivity of two independent Gaussians [27], we can directly sample $\mathbf{X}_t$ from the initial state $\mathbf{X}_0$:

$$Q(\mathbf{X}_t | \mathbf{X}_0) = \mathcal{N}(\mathbf{X}_t; \sqrt{\bar{\alpha}_t} \mathbf{X}_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad (8)$$

where $\bar{\alpha}_t = \prod_{t'=1}^t (1 - \beta_{t'})$ is the cumulative noise decay coefficient. In this process, the user and item representations are jointly

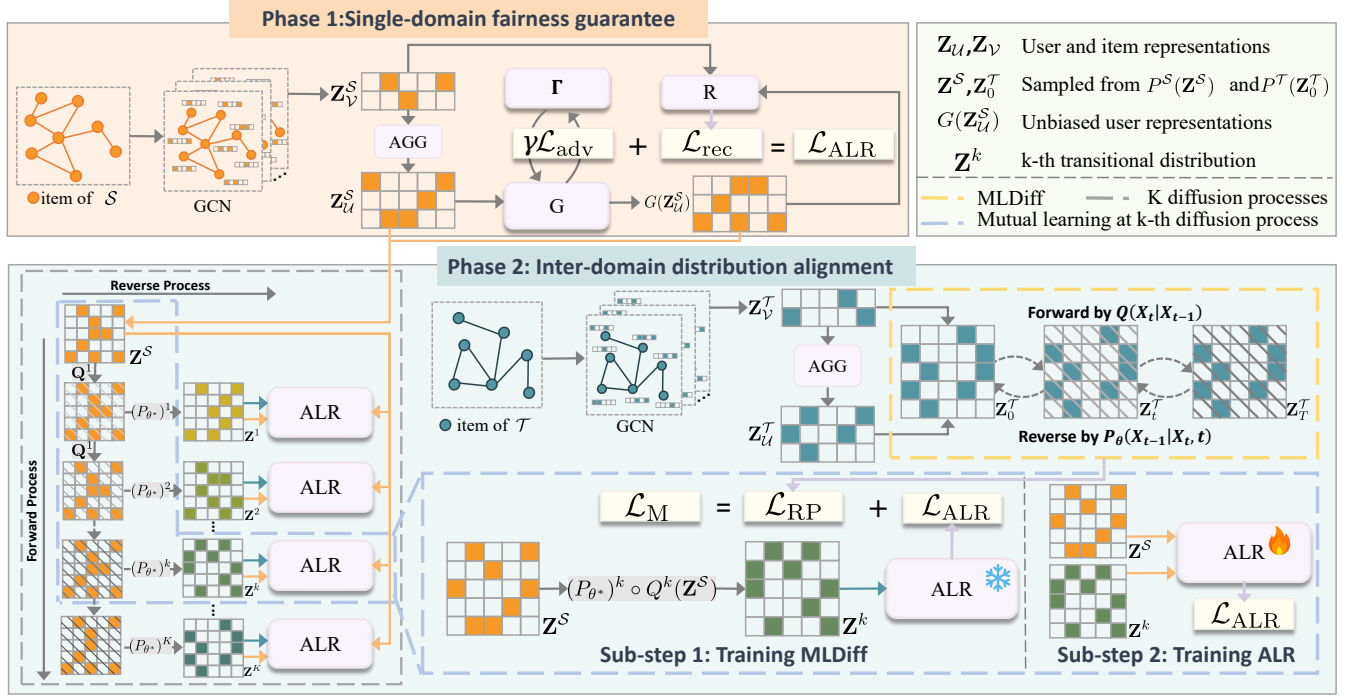


Figure 1: The training framework of FairCDR. FairCDR has two phases. In phase 1, we train ALR through adversarial learning for guaranteeing fairness on $\mathcal{S}$. In phase 2, we align the source and target domain distributions through K diffusion processes. In each process, we adopt mutual learning to train MLDiff and ALR alternately to enhance ALR’s capability of guaranteeing fairness and capturing user preferences on $\mathcal{T}$.

corrupted, thereby ensuring an efficient modeling of their interdependencies and potential relationships.

② *Reverse Process*. In this process, the model $P_\theta(\mathbf{X}_{t-1}|\mathbf{X}_t)$ learns to gradually denoise the corrupted state $\mathbf{X}_t$ to recover the initial state $\mathbf{X}_0$. Formally, it is defined as

$$P_\theta(\mathbf{X}_{t-1}|\mathbf{X}_t) = \mathcal{N}(\mathbf{X}_{t-1}; \mu_\theta(\mathbf{X}_t, t), \Sigma_\theta(\mathbf{X}_t, t)), \quad (9)$$

where $\mu_\theta(\mathbf{X}_t, t)$ and $\Sigma_\theta(\mathbf{X}_t, t)$ are the predicted mean and covariance, which are approximated by deep neural networks with learnable parameters θ . Following previous diffusion models [12, 27], the reverse process is optimized by the standard variational bound on the negative log-likelihood:

$$\mathcal{L}_{\text{RP}} = \mathbb{E}_q \left[-\log P(\mathbf{X}_T) - \sum_{t=1}^T \log \frac{P_\theta(\mathbf{X}_{t-1}|\mathbf{X}_t)}{Q(\mathbf{X}_t|\mathbf{X}_{t-1})} \right], \quad (10)$$

where $P(\mathbf{X}_T) = \mathcal{N}(\mathbf{X}_T; \mathbf{0}, \mathbf{I})$.

The learned P_{θ^*} is fixed as a distribution prior, and we use the diffusion process to align P^S with P_{θ^*} for approximating the alignment between P^S and P^T . The initial alignment state is $\mathbf{Z}^S \sim P^S(\mathbf{Z}^S)$ where $\mathbf{Z}^S = [G(\mathbf{Z}_U^S), \mathbf{Z}_V^S]$. At each step t , we can follow the similar calculations in Eq.(8-9). But ALR (described in Section 4.2.1) is only trained on the initial P^S , which cannot perceive the generation of the distribution of the target domain, resulting in poor performance on the knowledge transfer of fairness and user preferences.

To alleviate this problem, we mutually learn MLDiff and ALR during training. Concretely, we first construct a series of transitional representations between $\mathbf{Z}^S$ and $\mathbf{Z}^T$ through K diffusion processes, allowing ALR to train on these transitional representations to better perceive the distributional changes. In the k -th diffusion process, the k -th transitional representation is given as

$$\mathbf{Z}^k = (P_{\theta^*})^k \circ Q^k(\mathbf{Z}^S). \quad (11)$$

Here $Q^k(\cdot)$ represents the k -th diffusion operator which performs k standard forward processes. $(P_{\theta^*})^k$ represents the k -th denoise operator which performs k reverse processes. Due to the fact that (P_{θ^*}) has been optimized with $\mathbf{Z}^T$, so each reverse process can be guided by P_{θ^*} , enabling $Q^k(\mathbf{Z}^S)$ to denoise along the direction of $\mathbf{Z}^T$. Through this way, MLDiff generates a set of transitional representations, i.e., $\{\mathbf{Z}^k\}_{k=0}^K$ and $\mathbf{Z}^0 = \mathbf{Z}^S$.

Highlights. In traditional diffusion models, the numbers of forward process and of reverse process are fixed. In MLDiff, we dynamically increase k to progressively generate transitional representations closer to $\mathbf{Z}^T$. When k is small, the transitional representation remains close to $\mathbf{Z}^S$. For example, when $k = 1$, Gaussian noise is added to $\mathbf{Z}^S$ once and denoised once to generate $\mathbf{Z}^1$, so $\mathbf{Z}^1$ is close to $\mathbf{Z}^S$. As k increases from 1 to K , $\mathbf{Z}^k$ would steadily align $\mathbf{Z}^T$ by performing multiple diffusion and denoise operators.

Based on Z^k , we alternatively optimize MLDiff and ALR in each diffusion process. The objective of MLDiff is defined as

$$\mathcal{L}_M = \mathcal{L}_{RP} + \mathcal{L}_{ALR}. \quad (12)$$

$\mathcal{L}_{RP}$ is same with Eq.(10), enabling the generated representations to be close to Z^T , and $\mathcal{L}_{ALR}$ preserves fairness and user preferences during the alignment process as defined in Eq.(6). The objective of ALR is defined as Eq.(6), and its input now includes two parts: Z^S and Z^k . By introducing Z^k , ALR enhance its capability for guaranteeing fairness and capturing user preferences on $\mathcal{T}$.

Through mutual learning, ALR now is not only trained on the initial P^S , but also has been co-trained with MLDiff on $\{Z^k\}_{k=0}^K$ in K diffusion processes. Hence, ALR can progressively perceive the distributional changes, improving the knowledge transfer of fairness and user preferences for the target domain.

4.3 Model Analyses

Training Process. FairCDR is trained in two phases. In phase 1, we construct an item-item collaboration graph in the source domain to obtain $Z_{\mathcal{V}}^S$ and aggregate user interactions to derive $Z_{\mathcal{U}}^S$ in Eq.(2-4). We then apply adversarial learning to debias $Z_{\mathcal{U}}^S$ into $G(Z_{\mathcal{U}}^S)$ in Eq.(5) and train ALR via Eq.(6). In Phase 2, we use the same encoding method in Eq.(2-4) to generate $Z_{\mathcal{V}}^T$ and $Z_{\mathcal{U}}^T$ for the target domain. We concatenate $Z_{\mathcal{V}}^T$ and $Z_{\mathcal{U}}^T$ to train MLDiff for learning the distribution prior of $\mathcal{T}$ in Eq.(8-10). Under the guidance of this prior P_{θ^*} , we run K diffusion processes to gradually align Z^S with Z^T . During the k -th diffusion process, we generate the transitional distribution Z^k in Eq.(11) and mutually learn MLDiff and ALR via Eq.(12) and Eq.(6). This enables an effective transfer of fairness and user preferences. The pseudo-code of training FairCDR is provided in Appendix C.

Time Complexity. In phase 1, the time complexity of GCN for generating $Z_{\mathcal{V}}^S$ is $O(L|E|\hat{d} + L|\mathcal{V}^S|\hat{d}^2)$, where L is the layer number, $|E|$ is the number of edges and $\hat{d}$ is the dimension size. The step of the user feature aggregation has a time complexity of $O(|\mathcal{U}^S|m\hat{d})$, where m is the average number of items each user interacts with. The adversarial learning includes G and Γ . The time complexity of G for producing unbiased user representations depends on the used base recommendation model and we denote it as $O(\Omega)$. Γ is parameterized as a simple MLP and its time complexity is $O(|\mathcal{U}^S|\hat{d}^2)$. In phase 2, the main computational overhead comes from the mutual learning between MLDiff and ALR. The time complexity of this part is $O(|\mathcal{U}^S|\hat{d}^2 + \Omega)$. The overall time complexity of FairCDR is $O((L|E| + |\mathcal{U}^S|m)\hat{d} + (L|\mathcal{V}^S| + |\mathcal{U}^S|)\hat{d}^2 + \Omega)$. Based on the above analysis, the time complexity of FairCDR is primarily determined by the chosen base recommendation model, which is acceptable.

5 Experiments

5.1 Experimental Setup

5.1.1 Datasets. Due to the incomplete coverage of user attribute information in commonly used CDR datasets, here we construct three new real-world datasets which are derived from the Ali-CCP [28]. All of them are collected from Alimama's traffic logs². Following

²<https://tianchi.aliyun.com/dataset/408>

the previous setup [37], we use the feature "301" to divide these logs into three domain-specific datasets, i.e., D1, D2 and D3. Each dataset owns 13 user attributes, and we select gender as the sensitive attribute. The detailed information of datasets is provided in Appendix D. Based on these datasets, we generated four CDR datasets: D1→D2, D1→D3, D3→D1 and D3→D2. Here we do not use D2 as the source domain dataset, because the limited size of D2 restricts the effectiveness of cross-domain knowledge transfer.

5.1.2 Evaluation Protocols. According to previous approaches [40, 42], we employ a leave-one-out strategy for model evaluation: the most recent interaction of each user is treated as the test data, while the second-to-last interaction is used for validation, and the remaining data is used for training. For the measurement of recommendation performance, we use **Recall@N** and **NDCG@N** as evaluation metrics, where N is set to 10 and 30 respectively. For the measurement of fairness, we follow a standard setting used in [20, 21] and use the AUC score as the evaluation metric. The AUC score ranges from [0.5, 1], where *lower values* (values close to 0.5) indicate *better fairness*. A detailed explanation of fairness evaluation is provided in Appendix E.

5.1.3 Baselines. We compare FairCDR with the following three categories: 1) **Single-domain recommenders**: DuoRec³ [31], DCRec [40]⁴ and MAERec⁵ [42]. These methods perform recommendation by leveraging user interaction sequences within a single domain. 2) **Cross-domain recommenders**: RecGURU⁶ [18], C²DSR⁷ [2], SynCRec⁸ [30] and AMID⁹ [39]. They extend the recommendation task to a cross-domain setting by utilizing user behaviors from multiple domains to improve recommendation performance, but without considering fairness. 3) **Cross-domain recommenders with fairness penalty**: We introduce this new compared branch which combines cross-domain recommenders with fairness penalty for fully evaluating our model. Concretely, we add an adversarial loss [19] to recommendation loss as used in Eq.(5-6), i.e., w/L_{adv} to mitigate the influence of sensitive attributes, promoting fair representation learning for target domain users. The detailed description of compared baselines is provided in Appendix F.

5.1.4 Implementation Details. We use MAERec as the base recommendation model. For a fair comparison, the dimension size of users and items $\hat{d}$ is fixed as 128 for all models. The grid search is used for finding the optimal values of hyper-parameters. In particular, there are four main hyper-parameters: the learning rate lr , the harmonic factor in Eq.(6) γ , the number of diffusion processes K and the number of GCN layers L . The hyper-parameter intervals of them are listed correspondingly: $\{1e-4, 5e-4, 1e-3, 5e-3, 1e-2\}$, $\{0, 1, 2, 3, 4\}$, $\{0, 50, 100, 150, 200\}$ and $\{1, 2, 3, 4, 5\}$. In Section 5.5, we study the sensitivity of these hyper-parameters for our model. The configuration for running environment is provided in Appendix G.

³<https://github.com/RuihongQiu/DuoRec>

⁴<https://github.com/HKUDS/DCRec>

⁵<https://github.com/HKUDS/MAERec>

⁶<https://github.com/Chain123/RecGURU>

⁷<https://github.com/cjx96/C2DSR>

⁸<https://github.com/cpark88/SynCRec>

⁹<https://github.com/wujiangxu/amid>

Table 1: Comparison of recommendation performances and fairness between FairCDR and compared baselines.

Method \ Dataset	D1→D2					D1→D3				
	Recall↑		NDCG↑		AUC↓	Recall↑		NDCG↑		AUC↓
	@10	@30	@10	@30		@10	@30	@10	@30	
DuoRec	0.0145	0.0469	0.0061	0.0133	0.7832	0.1504	0.2405	0.0903	0.1115	0.6774
DCRec	0.0238	0.0832	0.0127	0.0262	0.8129	0.1612	0.2503	0.0976	0.1185	0.6934
MAERec	0.0284	0.1064	0.0129	0.0375	0.7666	0.1637	0.2706	0.0957	0.1209	0.6854
RecGURU	0.0701	0.2270	0.0405	0.0609	0.8733	0.2333	0.4013	0.1187	0.1584	0.7026
C²DSR	0.1120	0.2720	0.0614	0.0982	0.6584	0.2487	0.4803	0.1233	0.1782	0.5796
SyNCR	<u>0.1481</u>	0.2591	<u>0.0799</u>	<u>0.1061</u>	0.6435	0.2419	0.4403	<u>0.1412</u>	0.1835	0.6371
AMID	0.1477	0.2547	0.0610	0.0899	0.5809	<u>0.2324</u>	<u>0.4962</u>	0.1368	0.1970	0.6247
RecGURU w/L_{adv}	0.0498	0.1079	0.0232	0.0424	0.5312	0.1064	0.3041	0.0523	0.0944	0.5209
C²DSR w/L_{adv}	0.0960	0.1600	0.0455	0.0601	0.5243	0.1839	0.3985	0.0932	0.1430	0.5275
SyNCR w/L_{adv}	0.0831	0.1511	0.0437	0.0595	<u>0.5167</u>	0.0820	0.1714	0.0409	0.0597	0.5173
AMID w/L_{adv}	0.1026	0.2051	0.0510	0.0755	0.5155	0.2060	0.4121	0.0961	0.1563	<u>0.5165</u>
FairCDR (Ours)	0.1525	<u>0.2673</u>	0.0889	0.1156	0.5153	0.2524	0.5507	0.1474	<u>0.1795</u>	0.5132

Method \ Dataset	D3→D1					D3→D2				
	Recall↑		NDCG↑		AUC↓	Recall↑		NDCG↑		AUC↓
	@10	@30	@10	@30		@10	@30	@10	@30	
DuoRec	0.0773	0.1562	0.0399	0.0584	0.6492	0.0145	0.0469	0.0061	0.0133	0.7832
DCRec	0.0997	0.1658	0.0586	0.0741	0.6710	0.0238	0.0832	0.0127	0.0262	0.8129
MAERec	0.1135	0.2086	0.0616	0.0839	0.7163	0.0284	0.1064	0.0129	0.0375	0.7666
RecGURU	0.1587	0.3022	0.0654	0.1258	0.8301	0.0639	0.1616	0.0473	0.0594	0.7759
C²DSR	0.1992	0.3478	0.1012	<u>0.1365</u>	0.5894	0.1615	0.2874	0.0754	0.1128	0.6302
SyNCR	0.1702	0.3206	0.0947	0.1229	0.6231	<u>0.1764</u>	0.3093	<u>0.0912</u>	<u>0.1167</u>	0.6484
AMID	0.1426	<u>0.4112</u>	0.0667	0.1228	0.6027	0.1333	0.2813	0.0706	0.1050	0.6144
RecGURU w/L_{adv}	0.0833	0.2885	0.0375	0.0836	0.5309	0.0381	0.0850	0.0079	0.0235	0.5168
C²DSR w/L_{adv}	0.1421	0.2870	0.0713	0.1221	0.5244	0.1002	0.1783	0.0522	0.0706	0.5085
SyNCR w/L_{adv}	0.1289	0.2580	0.0649	0.0949	0.5159	0.1224	0.2021	0.0607	0.0797	<u>0.5085</u>
AMID w/L_{adv}	0.0973	0.2713	0.0449	0.0852	<u>0.5169</u>	0.0667	0.2000	0.0303	0.0587	0.5087
FairCDR (Ours)	<u>0.1963</u>	0.4492	<u>0.1003</u>	0.1383	0.5147	0.1789	<u>0.2917</u>	0.0934	0.1204	0.5043

5.2 Performance Comparison

Table 1 demonstrate the performance comparisons of topN recommendation and fairness on four CDR datasets. The best performance is in boldface and the second is underlined. From Table 1, we have the following observations:

- FairCDR demonstrates the excellent balance between recommendation performance and fairness for CDR. From the results across all datasets, its AUC remains consistently at the lowest level. At the same time, its recommendation performance ranks either the best or second best on all datasets, consistently outperforming baseline methods with fairness penalty. The most obvious improvement is that FairCDR achieves a 49% increase in Recall@10 compared to AMID w/L_{adv} on D1→D2.
- Single-domain recommenders perform substantially lower in recommendation performance than that of cross-domain recommenders due to the absence of source domain information. Furthermore, their high AUC scores indicate that

sensitive information is implicitly embedded in learned user representations.

- Cross-domain recommenders achieve good recommendation performance but exhibit poor fairness. For example, on D3→D2, SyNCR achieves comparable recommendation performance to FairCDR, but its AUC is 29% higher, indicating significantly lower fairness. By introducing adversarial loss, which is the third compared branch, cross-domain recommenders achieve better fairness on the target domain, but their recommendation performances are significantly compromised, e.g., Recall@10 of these baselines drops by 32% on average. The possible reason is that adversarial learning for fairness on sparse data tends to over-corrupt user behavior features. In contrast, FairCDR tackles data sparsity by transferring knowledge from the source domain to the target domain, maintaining fairness while preserving key user behavior patterns, thus achieving a better balance between recommendation performance and fairness.

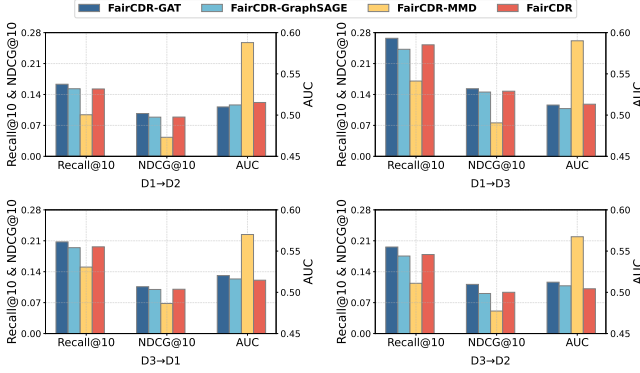


Figure 2: Performance comparison of model variants.

5.3 Model Variants

In this section, we try two classes of model variants of our model. The first one is to explore whether the proposed techniques are useful and reasonable, which includes 1) **FairCDR-GAT** adopts GAT [35] instead of the original used GCN to learn the structure information in constructed item-item collaboration graph; 2) **FairCDR-GraphSAGE** is similar to FairCDR-GAT, the only difference lies in that FairCDR-GraphSAGE adopts GraphSAGE [9] to learn the structure information. 3) **FairCDR-MMD** aligns the distributions of the source and target domains by minimizing the MMD loss [8], instead of using our proposed MLDiff. The second class is to explore the impact of the base model used in Eq.(6) in FairCDR. There are two model variants: 4) **FairCDR-DuoRec** and 5) **FairCDR-DCRec**, which respectively use DuoRec and DCRec as the base recommendation models. We perform these model variants on all datasets, and Recall@10, NDCG@10 and AUC are selected as evaluation metrics. The corresponding results are provided in Figure 2 and Table 2. From Figure 2, we can observe that:

- FairCDR-GAT performs better than FairCDR. We suppose the reason is that GAT assigns different weights to neighbors through an adaptive attention mechanism, better aggregating key information and improving recommendation results. However, since the improvement is not significant, considering computational overhead, we still adopt GCN. Additionally, FairCDR-GraphSAGE and FairCDR have similar performance.
- FairCDR-MMD performs worse than FairCDR in recommendation. This suggests that directly aligning domain distributions with MMD loss causes significant loss of semantic information, whereas MLDiff, with progressive alignment, enables more effective transfer of fairness and user preferences.

From Table 2 we can observe that FairCDR-DuoRec and FairCDR-DCRec achieve strong performance in both fairness and recommendation accuracy compared with DuoRec and DCRec, demonstrating FairCDR’s ability to enhance fairness and recommendation results across different base recommendation models.

Table 2: Impact of base recommendation models.

D1→D2	Recall@10	NDCG@10	AUC
DuoRec	0.0145	0.0061	0.7832
FairCDR-DuoRec	0.1330	0.0632	0.5087
DuoRec	0.0238	0.0127	0.8129
FairCDR-DCRec	0.1578	0.0712	0.5179
D1→D3	Recall@10	NDCG@10	AUC
DuoRec	0.1504	0.0903	0.6774
FairCDR-DuoRec	0.2330	0.1082	0.5077
DCRec	0.1612	0.0976	0.6934
FairCDR-DCRec	0.2482	0.1050	0.5102
D3→D1	Recall@10	NDCG@10	AUC
DuoRec	0.0773	0.0399	0.6492
FairCDR-DuoRec	0.1776	0.0904	0.5204
DCRec	0.0997	0.0586	0.6710
FairCDR-DCRec	0.1889	0.0847	0.5063
D3→D2	Recall@10	NDCG@10	AUC
DuoRec	0.0145	0.0061	0.7832
FairCDR-DuoRec	0.1683	0.0931	0.5079
DCRec	0.0238	0.0127	0.8129
FairCDR-DCRec	0.1704	0.0912	0.5123

5.4 Visualization of Transitional Distribution

To investigate whether MLDiff progressively aligns the distributions of the source and target domains, we visualize the transitional distributions generated by MLDiff and the target domain distribution on D3→D2. As shown in the Figure 3, the blue scatter points represent the transitional distribution sampled during the diffusion process (every 20 steps), while the orange scatter points represent the target domain distribution. Specifically, when $k=0$, the scatter plot illustrates the initial distribution states of the source and target domains. In each subfigure, 2,000 users are randomly sampled from both the transitional distribution and the target domain distribution. It can be observed that at the initial stage, the blue and orange point clusters are relatively far apart; as the diffusion steps increase (Figure 3b to Figure 3e), the blue points gradually move closer to the orange cluster. In the final stage (Figure 3f), the two distributions almost completely overlap, indicating that with the help of MLDiff, the transitional distribution successfully converges to and aligns with the target domain distribution.

5.5 Hyper-parameter Sensitivity

We investigate the hyper-parameter sensitivity with respect to four main hyper-parameters as described in Section 5.1.4. The experiment is conducted on D3→D2 and the empirical result is provided in Figure 4. From it, we have the following conclusions:

- FairCDR performs best at a learning rate of $1e-3$, so appropriately increasing the learning rate can improve the model’s recommendation quality. A smaller learning rate would result in slow convergence and inadequate learning, while a

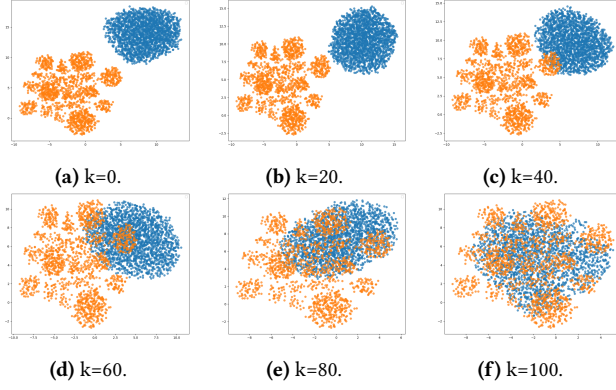


Figure 3: Transitional distributions at k -th diffusion process.

larger learning rate may cause gradient oscillations, slightly degrading performance.

- As the number of diffusion processes K increases, recommendation performance improves but saturates after $K > 100$. Initially, when $K = 0$, AUC is close to 0.5 due to insufficient learning of user preferences. As K increases, user preferences are learned more effectively, and sensitive attributes become more predictable. However, due to misalignment between the source and target domains, fairness in the target domain is not well maintained. As K increases further, the source and target domains align, improving fairness in the target domain.
- With the increase of γ , recommendation performance declines, while the AUC score decreases, indicating that the system is becoming increasingly fair.
- FairCDR achieves its best recommendation performance when the number of GCN layers is set to two. Increasing the number of GCN layers helps aggregate information and enhance recommendation performance. However, as the layers continue to grow, the model may suffer from the over-smoothing issue.

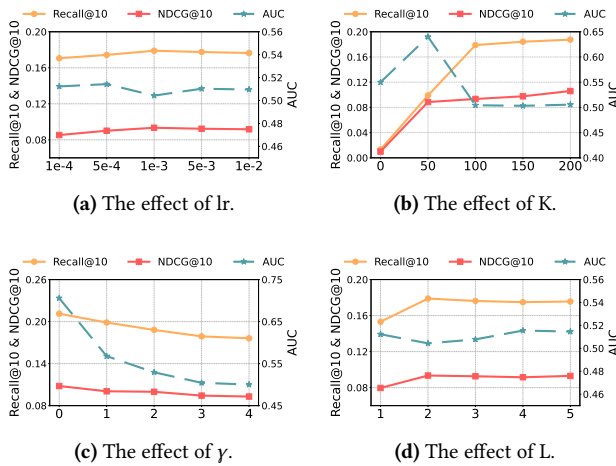


Figure 4: Hyper-parameter sensitivity of FairCDR.

6 Conclusion

In this paper, we explore the transferability of fairness in CDR to prevent biased results and ensure fair treatment of diverse user groups in the target domain. Our analysis shows that fairness can be effectively transferred if the source domain's fairness is ensured and the distributions of the source and target domains are well aligned. Based on this, we propose a novel framework, FairCDR, for transferring fairness and user preferences in CDR simultaneously. FairCDR first trains ALR to ensure fairness in the source domain, then aligns the distributions of the source and target domains through MLDiff. In the second phase, mutual learning of MLDiff and ALR allows ALR to gradually adapt to the target domain. Extensive experiments show that FairCDR outperforms existing strong baselines in terms of both fairness and recommendation quality.

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Appendix

A Notations

Table 3 presents the formal notations used in this paper.

Notation	Description
$\mathcal{S}, \mathcal{T}$	source and target domains
$\mathcal{D}^{\mathcal{S}}, \mathcal{D}^{\mathcal{T}}$	data of $\mathcal{S}$ and $\mathcal{T}$
$p^{\mathcal{S}}, p^{\mathcal{T}}$	distributions of $\mathcal{S}$ and $\mathcal{T}$
$\mathcal{U}^{\mathcal{S}}, \mathcal{V}^{\mathcal{S}}, \mathcal{E}^{\mathcal{S}}$	user set, item set, and corresponding interactions between users and items of $\mathcal{S}$
$\mathcal{U}^{\mathcal{T}}, \mathcal{V}^{\mathcal{T}}, \mathcal{E}^{\mathcal{T}}$	user set, item set, and corresponding interactions between users and items of $\mathcal{T}$
$\mathcal{U}^{\mathcal{O}}$	overlapping user set between $\mathcal{S}$ and $\mathcal{T}$
$\mathcal{X}$	user attributes shared by $\mathcal{S}$ and $\mathcal{T}$
$\tilde{\mathcal{A}}, \mathcal{A}$	sensitive and insensitive attributes
$\mathcal{G}^{\mathcal{S}}, \mathcal{G}^{\mathcal{T}}$	item-item collaboration graphs of $\mathcal{S}$ and of $\mathcal{T}$
$\mathbf{A}^{\mathcal{S}}$	adjacency matrix of $\mathcal{S}$
$\mathbf{Z}_{\mathcal{V}}^{\mathcal{S}}, \mathbf{Z}_{\mathcal{U}}^{\mathcal{S}}$	item and user representations of $\mathcal{S}$
$\mathbf{G}(\mathbf{Z}_{\mathcal{U}}^{\mathcal{S}})$	unbiased user representation for u of $\mathcal{S}$
$\mathbf{Z}_{\mathcal{V}}^{\mathcal{T}}, \mathbf{Z}_{\mathcal{U}}^{\mathcal{T}}$	item and user representations of $\mathcal{T}$
$\mathbf{Z}^{\mathcal{S}}, \mathbf{Z}^{\mathcal{T}}$	representation concatenated by $\mathbf{Z}_{\mathcal{V}}^{\mathcal{S}}$ and $\mathbf{Z}_{\mathcal{U}}^{\mathcal{S}}$, representation concatenated by $\mathbf{Z}_{\mathcal{V}}^{\mathcal{T}}$ and $\mathbf{Z}_{\mathcal{U}}^{\mathcal{T}}$
$\mathbf{Z}^k$	the k -th transitional representation

Table 3: Summary of used notations

B Proof of Theorem 4.1

To aid in the proof, we begin by introducing the definition of the Hausdorff distance.

Definition 3 (Hausdorff distance[1]). Given two point sets $A = \{a_1, a_2, \dots, a_n\}$ and $B = \{b_1, b_2, \dots, b_m\}$ in $\mathbb{E}^2$. The one-sided Hausdorff distance from A to B is defined as:

$$\tilde{\pi}_H(A, B) = \max_{a \in A} \min_{b \in B} \|a - b\|.$$

The bidirectional Hausdorff distance between A and B is then defined as:

$$\pi_H(A, B) = \max(\tilde{\pi}_H(A, B), \tilde{\pi}_H(B, A)).$$

The Hausdorff distance quantifies the greatest "outward" discrepancy between two sets by evaluating the closest points in each set relative to the other. It is a metric that is symmetric, non-negative, and adheres to the triangle inequality.

Next, we introduce several definitions that are instrumental for Theorem 4.1. Given a hypothesis space denoted by $\mathcal{H}$ and a true labeling function $f(\mathcal{Z}) : \mathcal{D} \rightarrow \mathcal{Y}$, we define the error of a hypothesis $G \in \mathcal{H}$ as $\epsilon^S(G, f) = \mathbb{E}_{\mathcal{Z} \sim \mathcal{D}^S} [|f(\mathcal{Z}) - G(\mathcal{Z})|]$, which represents the expected disagreement between the hypothesis G and the true labeling function f over the distribution $\mathcal{D}^S$.

The optimal joint hypothesis, which minimizes the combined error across both the source and target domains, is denoted by G^* . This hypothesis is defined as:

$$G^* = \arg \min_{G \in \mathcal{H}} (\epsilon^S(G, f) + \epsilon^T(G, f)), \quad (13)$$

where $\epsilon^T(G, f)$ denotes the error over the target distribution. Additionally, we introduce $\mathcal{H}$ -divergence between two probability distributions, as proposed in prior works [1, 33].

Definition 4 ($\mathcal{H}$ -divergence). The $\mathcal{H}$ -divergence is given by:

$$d_{\mathcal{H}}(\mathcal{D}, \mathcal{D}') = 2 \sup_{G \in \mathcal{H}} |P_{\mathcal{D}}[I(G)] - P_{\mathcal{D}'}[I(G)]|,$$

where $I(G)$ represents the set for which $G \in \mathcal{H}$ serves as the characteristic function, i.e., $Z \in I(G) \iff G(Z) = 1$.

An empirical approximation, $\hat{d}_{\mathcal{H}}(\mathcal{D}, \mathcal{D}')$, can be obtained by identifying a hypothesis h that maximizes the discrepancy between the samples drawn from $\mathcal{D}$ and $\mathcal{D}'$. This divergence serves as a useful measure for quantifying distributional differences, which is particularly relevant in domain adaptation scenarios, where the goal is to transition from the source domain to the target domain effectively.

Definition 5 (symmetric difference hypothesis space). The symmetric difference hypothesis space, denoted as $\mathcal{H}\Delta\mathcal{H}$, represents a set of hypotheses defined by:

$$G \in \mathcal{H}\Delta\mathcal{H} \iff G(Z) = h(Z) \oplus h'(Z), \quad \text{for some } h, h' \in \mathcal{H},$$

where $\oplus$ denotes the XOR operation.

This hypothesis space is particularly useful for capturing instances of disagreement between the hypothesis G and the target labeling function f .

PROOF SKETCH. Without loss of generality, we make an assumption that $\mathbb{E}[G_{\tilde{\mathcal{A}} \leftarrow \tilde{\mathbf{a}}}(u \mid \mathcal{A} = \mathbf{a}, \tilde{\mathcal{A}} = \tilde{\mathbf{a}})] < \mathbb{E}[G_{\tilde{\mathcal{A}} \leftarrow \tilde{\mathbf{a}}'}(u \mid \mathcal{A} = \mathbf{a}, \tilde{\mathcal{A}} = \tilde{\mathbf{a}})]$. Then we can rewrite $d^S(G)$ as follows:

$$\begin{aligned} d^S(G) &= \sum_{u \in \mathcal{U}^S} |\mathbb{E}[G_{\tilde{\mathcal{A}} \leftarrow \tilde{\mathbf{a}}}(u \mid \mathcal{A} = \mathbf{a}, \tilde{\mathcal{A}} = \tilde{\mathbf{a}})] \\ &\quad - \mathbb{E}[G_{\tilde{\mathcal{A}} \leftarrow \tilde{\mathbf{a}}'}(u \mid \mathcal{A} = \mathbf{a}, \tilde{\mathcal{A}} = \tilde{\mathbf{a}})]| \\ &= \epsilon_{\tilde{\mathcal{A}}}^S(G^*, f) + \epsilon_{\tilde{\mathcal{A}}}^S(1 - G^*, f) - 1, \end{aligned}$$

where the last line is derived from the fact that equal opportunity focuses solely on the error associated with the false data points. Then we estimate the upper bound of $d_{\text{CF}_T}(G)$ as:

$$\begin{aligned} &d^T(G) \\ &= \epsilon_{\tilde{\mathcal{A}}}^T(G^*, f) + \epsilon_{\tilde{\mathcal{A}}}^T(G, G^*) + \epsilon_{\tilde{\mathcal{A}}}^S(G^*, f) + \epsilon_{\tilde{\mathcal{A}}}^S(1 - G, G^*) - 1 \\ &\leq \epsilon_{\tilde{\mathcal{A}}}^T(G^*, f) + \epsilon_{\tilde{\mathcal{A}}}^T(G, G^*) + \frac{1}{2} d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{U}^{S_{\tilde{\mathcal{A}}}}, \mathcal{U}^{T_{\tilde{\mathcal{A}}}}) + \epsilon_{\tilde{\mathcal{A}}}^T(G^*, f) \\ &\quad + \epsilon_{\tilde{\mathcal{A}}}^S(1 - G, G^*) - 1 \\ &\leq \epsilon_{\tilde{\mathcal{A}}}^T(G^*, f) + \epsilon_{\tilde{\mathcal{A}}}^S(G, f) + \frac{1}{2} d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{U}^{S_{\tilde{\mathcal{A}}}}, \mathcal{U}^{T_{\tilde{\mathcal{A}}}}) + \epsilon_{\tilde{\mathcal{A}}}^T(G^*, f) \\ &\quad + \epsilon_{\tilde{\mathcal{A}}}^S(1 - G, f) - 1 \\ &= d^S(G) + \frac{1}{2} \hat{d}_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{U}^{S_{\tilde{\mathcal{A}}}}, \mathcal{U}^{T_{\tilde{\mathcal{A}}}}) + \lambda_{\tilde{\mathcal{A}}}(\mathcal{G}^*, f) \\ &\leq d^S(G) + \frac{1}{2} d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{U}^{S_{\tilde{\mathcal{A}}}}, \mathcal{U}^{T_{\tilde{\mathcal{A}}}}) + \pi_H(\mathcal{U}^{S_{\tilde{\mathcal{A}}}}, \mathcal{U}^{T_{\tilde{\mathcal{A}}}}) \\ &\quad + 8 \sqrt{\frac{2d \log(2m') + \log(\frac{2}{\delta})}{m'}} + \lambda_{\tilde{\mathcal{A}}}(\mathcal{G}^*, f) \\ &\Leftrightarrow \\ &\epsilon \leq \frac{1}{2} d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{U}^{S_{\tilde{\mathcal{A}}}}, \mathcal{U}^{T_{\tilde{\mathcal{A}}}}) + \pi_H(\mathcal{U}^{S_{\tilde{\mathcal{A}}}}, \mathcal{U}^{T_{\tilde{\mathcal{A}}}}) + \lambda^*, \end{aligned}$$

where $\lambda^* = \lambda(d, m', \delta) + \lambda_{\tilde{\mathcal{A}}}(\mathcal{G}^*, f)$ and we complete the proof. $\square$

C The pseudo-code of FairCDR

Algorithm 1 shows the training procedures of FairCDR.

D Details of Datasets

Table 4 shows the statistical information of three datasets.

Table 4: The statistics of used datasets.

	#Users	#Items	#Clicks	Density
D1	82k	286k	1.3M	5.54×10^{-5}
D2	2k	20k	28k	7.00×10^{-4}
D3	135k	352k	2.0M	4.21×10^{-5}

E Fairness Evaluation

We follow previous representative methods [20, 21] to achieve fairness evaluation. To be specific, we additionally train an attacker model to infer users' sensitive attributes, which has the same structure and capacity as the discriminator Γ . When the training of FairCDR has been completed, we use 80% of the target domain users as the training set and train the attacker with their representations learned from FairCDR, reserving 20% for testing. The performance of the attacker's inference, i.e., the AUC score, can reflect the fairness of the model. A lower AUC score implies that the attacker finds it more difficult to distinguish users with different sensitive attributes, indicating stronger fairness against sensitive attribute inference. When the AUC score approaches 0.5, the attacker can hardly leverage user representations for effective identification, signifying an ideal fairness.

Algorithm 1: Training procedures of FairCDR.

Require: The number of diffusion processes K , source and target domain data $\mathcal{D}^S$ and $\mathcal{D}^T$.

Ensure : Learned model parameters of ALR are used to transfer knowledge of fairness and user preferences to the target domain.

- 1 **Phase 1: Single-domain fairness guarantee**
- 2 Construct an item-item collaboration graph on $\mathcal{S}$
- 3 $Z_{\mathcal{V}}^S, Z_{\mathcal{U}}^S \leftarrow \text{Eq.(2-4)}$; // Initial user and item representations on $\mathcal{S}$
- 4 $\mathcal{L}_{\text{adv}} \leftarrow \text{Eq.(5)}$; // Adversarial learning
- 5 $\mathcal{L}_{\text{ALR}} \leftarrow \text{Eq.(6)}$; // Loss of ALR on $\mathcal{S}$
- 6 **Phase 2: Inter-domain distribution alignment**
- 7 Construct an item-item collaboration graph on $\mathcal{T}$
- 8 $Z_{\mathcal{V}}^T, Z_{\mathcal{U}}^T \leftarrow \text{Eq.(2-4)}$; // Initial user and item representations on $\mathcal{T}$
- 9 **for** $t = 1, \dots, K$ **do**
- 10 /* **sub-step 1: Training MLDiff** */
- 11 Freeze ALR;
- 12 $Z^k \leftarrow \text{Eq.(11)}$; // Transitional representation
- 13 $\mathcal{L}_M \leftarrow \text{Eq.(12)}$; // Loss of MLDiff
- 14 /* **sub-step 2: Training ALR** */
- 15 Freeze MLDiff;
- 16 $\mathcal{L}_{\text{ALR}} \leftarrow \text{Eq.(6)}$; // Update ALR on $\{Z^S, Z^k\}$

F Baseline Description

We compare FairCDR with the following strong baselines:

- **DuoRec** [31] uses contrastive learning to address representation degeneration in sequential recommendation, improving recommendation accuracy and representation quality.

- **DCRec** [40] leverages debiasing contrastive learning to unify sequence pattern encoding and global collaborative modeling for enhancing recommendation performance.
- **MAERec** [42] proposes a sequential recommendation model based on a graph-masked autoencoder, which can adaptively extract global item transition information to improve recommendation performance.
- **RecGURU** [18] proposes an adversarial learning-based CDR framework which can enhance recommendation results by generating generalized user representations.
- **C²DSR** [2] combines graph neural networks with sequential attention to jointly learn single-domain and cross-domain user preferences while enhancing cross-domain relevance through contrastive learning.
- **SyNCR** [30] is designed for adaptive optimization between single-domain and cross-domain sequential recommendation, enhancing information transfer and recommendation results.
- **AMID** [39] handles data distribution differences between source and target domains through a multi-interest module and dual robust estimation to improve recommendation results.

For the third compared branch, i.e., cross-domain recommenders with fairness penalty, we add an adversarial loss [19] to the recommendation loss of RecGURU, C²DSR, SyNCR and AMID to achieve fair recommendation in the target domain.

G Running Environment

The experiments are conducted on a Linux server with Intel(R) Xeon(R) CPU E5-2620 v4 @ 2.20GHz, 128G RAM and NVIDIA Tesla V100. The operating system is Ubuntu 16.04.